

11. (2.5 points)

An actuary is building a stochastic chain ladder model and is considering the following distributions:

- Over-Dispersed Poisson
- Over-Dispersed Negative Binomial
- Normal

Given the following information:

- Actual accident year 2015 losses at 12 months: \$50,000
- Estimated chain-ladder accident year 2015 losses at 24 months: \$75,000

a. (1.5 point)

For each of the following distributions, calculate the variance of the accident year 2015 losses at 24 months.

- Over-dispersed Poisson model with $\phi = 1.5$.
- Over-dispersed negative binomial model with $\phi = 1.25$.
- Normally distributed model with $\phi = 1.75$.

b. (0.5 point)

The actuary wants to use a model where the connection to the chain ladder method is immediately apparent. Identify and briefly explain which of the three models under consideration would achieve this.

c. (0.5 point)

The loss development triangle being used has a column of incremental values with a negative sum. The actuary wants to use a model that does not require adjustments to the data. Identify and briefly explain which of the three models under consideration would achieve this.

CONTINUED ON NEXT PAGE

EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 11	
TOTAL POINT VALUE: 2.5	LEARNING OBJECTIVE: A6: Calculate the mean and prediction error of a reserve given an underlying statistical model.
SAMPLE ANSWERS	
Part a: 1.5 points	
<i>Sample Answer 1</i> i) ODP: $x_i y_i \phi = 75000(1/1 - 1/1.5) \times 1.5 = 37,500$ ii) Neg bin: $\phi \lambda_i (\lambda_i - 1) \times D_{i,j-1} = 1.25(1.5)(1.5 - 1) \times 50,000 = 46,875$ iii) Norm: $\phi \times D_{i,j-1} = 1.75(50,000) = 87,500$ <i>Sample Answer 2</i> i) $\lambda = 1.5$ Variance = $1.5 \times 25,000 = 37,500$ ii) Variance = $50,000 \times 1.5 \times .5 \times 1.25 = 46,876$ iii) Variance = $1.75 \times 50,000 = 87,500$	
Part b: 0.5 point	
<i>Sample Answer 1</i> The negative-Binomial would achieve the goal as its format is most closely matched to the chain ladder formula. <i>Sample Answer 2</i> ODNB would achieve this since Expected value = $\phi \times (\lambda_i - 1) \times D_{i,j-1}$  Link factor for the incremental losses  cumulative	
Part c: 0.5 point	
<i>Sample Answer 1</i> The normal model would achieve this as both ODP and NB model cannot work where there is negative sum of increments due to variance constraints. <i>Sample Answer 2</i> Normal since this continuous pdf has a support of $(-\infty, +\infty)$ (includes negative)	
EXAMINER'S REPORT	
Part a	
Candidates were expected to know the three different variance formulas for the Over-Dispersed Poisson, the Over-Dispersed Negative Binomial, and the Normal distributions described in the Verrall paper. A minority of the candidates received full credit (all three formulas and correct numerical answers) for this part, with the remaining candidates roughly evenly split between knowing 0, 1, or 2 formulas and then calculating the right answer.	

EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

Part b

Candidates were expected to know that the mean formulas of two of the three distributions, the Over-Dispersed Negative Binomial and the Normal, are directly comparable to the chain ladder's mean formula (because they both have the form of a LDF times a reported-to-date value).

Candidates generally struggled with this part, with the common error being naming the Over-Dispersed Poisson as the appropriate distribution. Several common erroneous justifications of the ODP (or other models) were:

- Same reserve estimates as chain ladder (this is true, but all models have the same reserve estimates hence this can't be used as a justification to pick one model over the others).
- The model with the highest variance.
- Variance matches chain ladder.

Part c

Candidates were expected to know that only the Normal distribution could easily (i.e., without adjustment to the data) handle negative incremental values because it is the only distribution of the three with support over negative values.

Candidates generally did very well on this part, though some candidates named one of the other models or left the question blank.